Supplementary Material for "DFT-informed stacked ensembles for experimental composition–property prediction: two pre-registered, fully nested evaluations on Matbench"

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S1. Pre-registration, amendments and the post-hoc ablation

The analysis plan (`ANALYSIS_PLAN.md`) and SHA-256 hashes of all code were committed before each scoring of the official test folds.

- **Original freeze** (`results/freeze_hashes_original_plan.txt`). Under it, the design was fixed on inner CV of fold 0's training rows. A run under this plan produced test predictions that were never scored; they are archived unscored.
- **Amendment 1 = run 1** (`results/freeze_hashes.txt`, plan Section 8). Written before any scoring. It moved the choice among the explored design configurations inside every official training fold (Section S3).
- **Amendment 2 = run 2** (`results/freeze_hashes_amendment2.txt`, plan Sections 9–11). Written after run 1 had been scored.

  - It changed the expert pool, the features (including the disclosed DFT block) and the correction gate.
  - It was frozen before any run-2 test prediction, and scored once.
  - The design evidence (fold 0 training rows only) is in `results/design_a2/DESIGN_LOG_A2.md`.

- **Post-hoc ablation** (plan Section 12). Written after run 2 had been scored and not used for any selection. It re-runs the frozen run-2 code without the DFT block (all tasks) or without the hurdle experts (expt_gap).

- **Deviations** are listed in plan Sections 7 and 11.

The leaders' per-sample predictions come from the public `matbench` repository (`benchmarks/matbench_v0.1_{TPOT,d`

S2. Run-1 methods: local correction, fold-matched variant and hard labels

0.1 Novelty-damped local OOF residual correction

With OOF residuals $r_j = y_j - s^{\text{oof}}(x_j)$, the corrected prediction for a query q is

$$\hat{y}(q) = s(q) + \alpha \, \frac{\sum_{j \in \mathcal{N}_k(q)} w_{qj} \, r_j}{\sum_{j \in \mathcal{N}_k(q)} w_{qj} + \kappa}, \qquad w_{qj} = \exp\big(- (d(q,j)/h)^2 \big), \tag{1}$$

where $\mathcal{N}_k(q)$ are the k nearest training compositions under a composition distance d (Euclidean on standardised Magpie vectors, or L1 on element fractions). The pseudo-count κ provides the novelty damping: when all neighbours are far, $\sum w$ is small and the correction vanishes. For classification the shift is applied on the logit scale with probability residuals. The tuple $(d, k, h, \kappa, \alpha)$ is selected on the training rows only, by scoring the corrected OOF predictions (leave-self-out neighbour sets) with the task metric over a fixed grid: $k \in \{5, 10, 20, 40\}$, h a multiple $\{0.5, 1, 2, 4, 8, 16\}$ of the median nearest-neighbour distance, $\kappa \in \{0, 0.1, 0.5, 2\}$, and $\alpha \in \{0.25, 0.5, 0.75, 1\}$ (regression) or $\{0.1, 0.25, 0.5, 1, 2, 4\}$ (classification). The uncorrected stack ($\alpha = 0$) is always a candidate and wins ties, so the procedure may decline to correct.

0.2 Fold-matched correction

The deployed stack $s = F^{\text{full}}$ was trained on the very neighbours j whose OOF residuals Eq. (1) borrows. Writing $F^{(-m)}$ for the stack whose experts excluded inner fold P_m, the naive transfer decomposes as $r_j = [r_j + F^{(-m(j))}(q) - F^{\text{full}}(q)] + [F^{\text{full}}(q) - F^{(-m(j))}(q)]$. The second bracket, the *absorption term*, measures how far the refit has already moved toward the neighbour, and the naive correction counts it a second time; for interpolating experts and near-duplicate neighbours it can be as large as r_j itself. The fold-matched variant transfers only the first bracket, $T_{qj} = r_j + F^{(-m(j))}(q) - F^{\text{full}}(q)$, in place of r_j in Eq. (1). Its selection uses deployment-matched training instances: for a training row $i \in P_m$ and a neighbour $j \in P_{m'}$ ($m' \neq m$) the transfer is computed from the pair-excluded refit $F^{(-m,-m')}$, which saw neither i nor j; neighbours in the same inner fold use r_j directly. This requires $J(J-1)/2 = 10$ extra fits per expert and fold, with the stack weights held fixed. The variant was proposed by a structured ideation pipeline (ResearchStudio IdeaSpark), whose coherence check verified the algebraic decomposition on synthetic instances; we pre-registered it as a secondary arm, not as the primary method. Relative to that proposal we simplified the damping map to the same (α, κ) family as Eq. (1), used the same two distances instead of the element mover's distance, and did not add TabPFN or MODNet experts, for reasons of CPU cost.

0.3 Hard labels for classification

Most classification entries on the Matbench leaderboard report a "ROC-AUC" equal to their balanced accuracy, which is what `matbench` computes when hard labels rather than probabilities are recorded. For a like-for-like comparison we therefore also report balanced accuracy at a threshold chosen inside each training fold (maximising balanced accuracy on the arm's training OOF predictions), alongside the probability ROC-AUC.

S3. Per-fold design selection (run 1)

Inside each official training fold, every candidate was cross-fitted and the one with the best cross-fitted OOF score of the uncorrected stack was kept.

Regression candidates:

- A: base features, L2 stack;
- B: base features, L1 stack, plus an L1-objective LightGBM expert;
- D: extra features, L1 stack;
- C: as D, plus the L1-LightGBM expert.

Classification candidates: A (base features) and C (extra features).

Table 1: Per-fold design selection in run 1.

Task	Fold	Selected design	Candidate stack OOF scores
steels	0	B_base_l1	A=84.3708, B=83.1777, D=85.4401, C=85.4401
steels	1	D_extra_l1	A=90.1208, B=89.8331, D=89.2622, C=89.2622
steels	2	A_base_l2	A=92.3679, B=94.5285, D=93.8040, C=93.8040
steels	3	C_extra_l1_lgbml1	A=92.6136, B=92.0424, D=91.4351, C=91.4351
steels	4	B_base_l1	A=88.8951, B=88.8950, D=89.2953, C=89.2953
expt_gap	0	D_extra_l1	A=0.4159, B=0.4121, D=0.3408, C=0.3408
expt_gap	1	D_extra_l1	A=0.4192, B=0.4180, D=0.3408, C=0.3409
expt_gap	2	D_extra_l1	A=0.4067, B=0.4021, D=0.3411, C=0.3411
expt_gap	3	D_extra_l1	A=0.4175, B=0.4185, D=0.3451, C=0.3454
expt_gap	4	D_extra_l1	A=0.4115, B=0.4062, D=0.3385, C=0.3385
expt_is_metal	0	C_extra	A=0.9739, C=0.9819
expt_is_metal	1	C_extra	A=0.9733, C=0.9808
expt_is_metal	2	C_extra	A=0.9746, C=0.9813
expt_is_metal	3	C_extra	A=0.9725, C=0.9795
expt_is_metal	4	C_extra	A=0.9743, C=0.9812
glass	0	C_extra	A=0.9458, C=0.9461
glass	1	A_base	A=0.9521, C=0.9520
glass	2	C_extra	A=0.9515, C=0.9523
glass	3	C_extra	A=0.9508, C=0.9508
glass	4	C_extra	A=0.9449, C=0.9453

Table 2: Every arm on the four Matbench composition tasks (official 5-fold CV; mean ± s.d. over folds). Regression: MAE (lower is better); classification: ROC-AUC from probabilities (higher is better). Single-expert rows are means over the folds in which the expert was part of the selected configuration.

Arm	steels MAE (MPa)	expt_gap MAE (eV)	expt_is_metal ROC-AUC	glass ROC-AUC
Best single (inner CV)	83.41 ± 10.86	0.3259 ± 0.0130	0.9819 ± 0.0015	0.9548 ± 0.0097
Equal-weight mean	102.46 ± 20.11	0.4007 ± 0.0242	0.9811 ± 0.0027	0.9458 ± 0.0116
Simplex stack	83.12 ± 10.89	0.3264 ± 0.0139	0.9825 ± 0.0017	0.9567 ± 0.0091
Stack + local corr. (naive)	82.80 ± 11.65	0.3240 ± 0.0140	0.9825 ± 0.0018	0.9567 ± 0.0091
ablation: Magpie space only	83.53 ± 11.30	0.3235 ± 0.0139	0.9825 ± 0.0017	0.9567 ± 0.0091
ablation: fraction space only	82.80 ± 11.65	0.3235 ± 0.0142	0.9825 ± 0.0018	0.9567 ± 0.0091
Stack + fold-matched corr.	82.02 ± 9.12	0.3276 ± 0.0149	0.9826 ± 0.0017	0.9559 ± 0.0094
Best single + local corr.	82.69 ± 11.60	0.3239 ± 0.0133	0.9819 ± 0.0015	0.9548 ± 0.0097
Expert et_all	83.41	0.3259	0.9809	0.9548
Expert krr_frac	160.38	0.5613	–	–
Expert krr_magpie	109.37	0.4840	–	–
Expert lgbm_all	88.00	0.3425	0.9819	0.9519
Expert lgbm_l1_all	93.97	–	–	–
Expert mlp_all	156.17	0.4611	0.9702	0.9055
Expert rf_all	96.62	0.3645	0.9808	0.9516
Expert svc_frac	–	–	0.9472	0.8735
Expert svc_magpie	–	–	0.9566	0.9009

S4 Results I: the transferred local correction (Amendment 1)

S4.1 All arms on the official folds

Table 2 lists every arm on every task (mean ± standard deviation over the five official folds), and Table 3 the pre-registered tests. The equal-weight mean of the experts is clearly the worst ensemble on all tasks; the simplex stack never loses to it. Against the best single expert chosen by inner cross-validation, the stack is better on glass (+0.20% ROC-AUC, $p_{\text{boot}} = 0.002$, 5/5 folds) and is within noise on the other three tasks (steels -0.35%, expt_gap +0.13%, expt_is_metal +0.06%). The extremely randomised trees expert carries most of the stack weight on the two regression tasks, and LightGBM and extra trees share it on the classification tasks (Table 7).

S4.2 Primary endpoint: does the local correction transfer?

The correction improved 1 of the four tasks under the pre-registered rule. On expt_gap the corrected stack reached an MAE of 0.3240 eV against 0.3264 eV for the uncorrected stack ($\Delta = -0.00232$ eV, -0.71%), with every one of the five folds improved and $p_{\text{Holm}} = 0.005$ (Nadeau–Bengio $p = 0.011$). The effect is real but small: about 0.7% of the MAE. On steels the change was -0.39% ($p_{\text{Holm}} = 1.000$, 3/5 folds improved), indistinguishable from zero given a fold-to-fold standard deviation of about 10.89 MPa. On the two classification tasks the inner-CV selection almost never switched the correction on (glass: 0/5 folds; expt_is_metal: 1/5), so the corrected and uncorrected stacks are identical or nearly so (Δ ROC-AUC $= -0.00000$ and $+0.00000$). In short, the correction that was confirmed in our chemistry study transfers only weakly: it gives a consistent sub-percent gain on the largest regression task, nothing measurable on steels, and is (correctly) declined by its own selection procedure on the classification tasks.

Table 3: Pre-registered primary test (naive local correction vs. uncorrected stack) and key secondary comparisons. Δ is the mean over folds of the metric difference (MAE for regression, where negative favours A; ROC-AUC for classification, where positive favours A). CI and p_{boot}: paired bootstrap over test points within folds ($B = 10^4$); p_{Holm}: Holm correction over the four primary tests; p_{NB}: Nadeau–Bengio corrected resampled t-test over the five folds.

Task	Comparison (A vs B)	Δ = A−B (rel.)	95% CI	p_{boot}	p_{Holm}	folds A better	p_{NB}
steels	naive local vs stack (primary)	-0.3202 (-0.39%)	[-2.15, +1.5]	0.720	1.000	3/5	0.896
steels	naive local vs best single	-0.6135 (-0.74%)	[-2.63, +1.34]	0.538	–	3/5	0.816
steels	stack vs best single	-0.2933 (-0.35%)	[-1.65, +1.08]	0.660	–	2/5	0.417
steels	fold-matched vs stack	-1.097 (-1.32%)	[-4.32, +2.13]	0.519	–	2/5	0.817
steels	fold-matched vs naive local	-0.7765 (-0.94%)	[-3.45, +1.91]	0.583	–	3/5	0.812
expt_gap	naive local vs stack (primary)	-0.00232 (-0.71%)	[-0.0037, -0.000942]	0.001	0.005	5/5	0.011
expt_gap	naive local vs best single	-0.001895 (-0.58%)	[-0.00391, +0.000155]	0.071	–	4/5	0.108
expt_gap	stack vs best single	+0.0004248 (+0.13%)	[-0.000988, +0.00181]	0.566	–	3/5	0.695
expt_gap	fold-matched vs stack	+0.001241 (+0.38%)	[-0.00145, +0.00418]	0.390	–	2/5	0.516
expt_gap	fold-matched vs naive local	+0.00356 (+1.10%)	[+0.0012, +0.00602]	0.005	–	0/5	0.096
expt_is_metal	naive local vs stack (primary)	-3.305e-06 (-0.00%)	[-4.88e-05, +4.3e-05]	0.899	1.000	0/5	0.541
expt_is_metal	naive local vs best single	+0.0005853 (+0.06%)	[-0.000504, +0.0017]	0.301	–	4/5	0.348
expt_is_metal	stack vs best single	+0.0005886 (+0.06%)	[-0.000507, +0.0017]	0.275	–	4/5	0.345
expt_is_metal	fold-matched vs stack	+0.0001571 (+0.02%)	[-0.000148, +0.00047]	0.320	–	3/5	0.577
expt_is_metal	fold-matched vs naive local	+0.0001604 (+0.02%)	[-0.000128, +0.000458]	0.277	–	3/5	0.566
glass	naive local vs stack (primary)	+0 (+0.00%)	[+0, +0]	1.000	1.000	0/5	1.000
glass	naive local vs best single	+0.001897 (+0.20%)	[+0.000704, +0.00314]	0.002	–	5/5	0.141
glass	stack vs best single	+0.001897 (+0.20%)	[+0.000689, +0.00314]	0.002	–	5/5	0.141
glass	fold-matched vs stack	-0.0008173 (-0.09%)	[-0.00194, +0.000315]	0.153	–	1/5	0.247
glass	fold-matched vs naive local	-0.0008173 (-0.09%)	[-0.00192, +0.000273]	0.149	–	1/5	0.247

S4.3 Where the correction helps and hurts

Fig. 1 stratifies the test errors by the distance from each test composition to its nearest training composition. On both regression tasks the naive correction *hurt* in the nearest-support quartile (expt_gap +0.9%, steels +6.8% MAE relative to the stack) and helped in the other three (expt_gap -1.0%, -1.1%, -0.9%). That is the pattern the fold-matched analysis predicts: for a near-duplicate neighbour the deployed stack has already absorbed the neighbour's error, so adding its OOF residual counts it twice. The fold-matched variant, however, did not repair it: on expt_gap it was worse than the naive correction overall (+1.10% MAE, $p_{\text{boot}} = 0.005$, 0/5 folds better) and also worse in the nearest quartile (+2.0%). On steels it gave the lowest mean MAE of all arms (82.02 MPa) but with no detectable difference from the stack (-1.32%, $p_{\text{boot}} = 0.519$). On the classification tasks it was switched on in every fold but changed ROC-AUC by less than 0.1% in either direction.

S4.4 Ablations and controls

Restricting the neighbour search to one composition space gave essentially the same gain on expt_gap (Magpie -0.86%, element fractions -0.88%), whereas on steels only the element-fraction space was selected in practice and the Magpie-only variant was slightly worse than the stack (+0.50%, 0/5 folds better). Applying the correction to the best single expert instead of the stack ("correction without stacking") improved expt_gap by -0.63% relative to that expert ($p_{\text{boot}} = 0.002$, 5/5 folds), so on this task the gain does not depend on stacking. The label-permutation control did not behave as a clean null: with labels permuted within inner folds the selection still switched a correction on in every fold of every task (Table 4), because shrinking predictions toward a neighbourhood mean lowers the training loss against permuted labels too. We pre-registered this control as diagnostic only, and it shows that a non-zero selected strength is not by itself evidence of local structure.

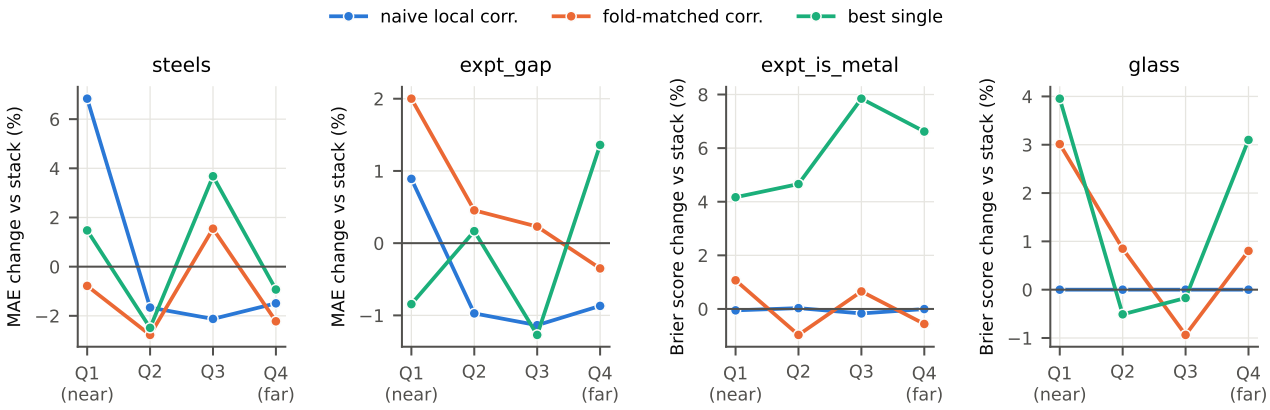


Figure 1: Error change relative to the uncorrected stack, by support quartile (distance from the test composition to its nearest training composition; bin edges are the training rows' leave-one-out quartiles in each fold). Regression: MAE; classification: Brier score. Negative is better.

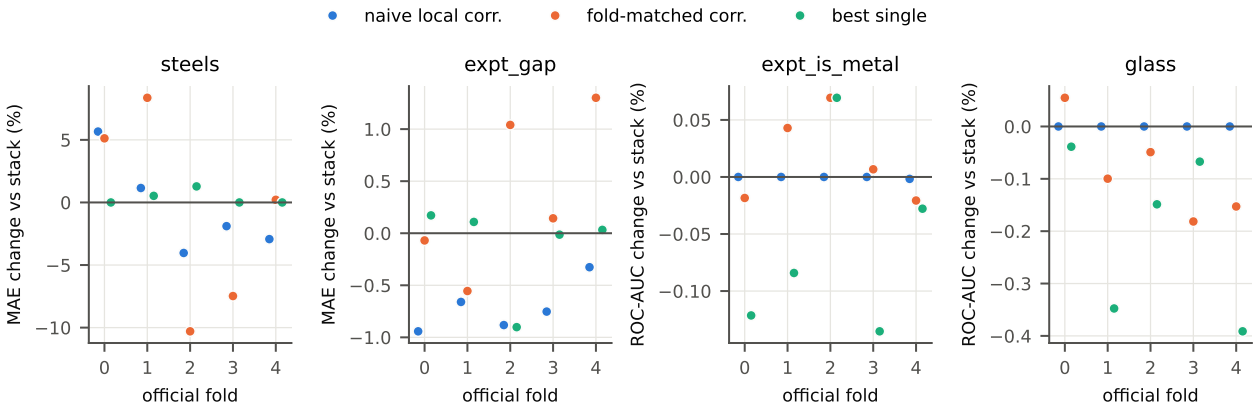


Figure 2: Per-fold change of the test metric relative to the uncorrected stack for the naive and fold-matched corrections and the best single expert (selected by inner CV). For classification a positive change in ROC-AUC is better.

S4.5 Comparison with the Matbench leaderboard

Table 5 places the pre-registered primary arm on the leaderboard snapshot of 18 October 2026. For the classification tasks we compare balanced accuracy at the inner-CV threshold with the board's balanced-accuracy column, because the board's "ROC-AUC" values of the label-based entries equal their balanced accuracy. The method would rank 3 of 12 on steels (behind TPOT-Mat and AutoML-Mat, ahead of MODNet v0.1.12), 2 of 13 on expt_gap (behind Darwin, ahead of the CrabNet and MODNet entries), 2 of 8 on expt_is_metal and 2 of 8 on glass. It does not beat the best entry on any task. On expt_gap and glass the gaps to the leaders are large (0.3240 vs. 0.2865 eV; balanced accuracy 0.8841 vs. 0.9603), and on expt_is_metal it is 0.023 in balanced accuracy. Our probability ROC-AUCs (0.9825 on expt_is_metal, 0.9567 on glass) are not comparable with the label-based board values and are not ranked.

S4.6 Calibration and compute

Stacking improved the Brier score and the expected calibration error over the best single expert on both classification tasks, and the corrections left them essentially unchanged (Table 6). On 24 CPU cores

Table 4: Number of official folds (of 5) in which the inner-CV selection switched each correction on, and the same for the label-permutation control.

Task	naive corr. on	fold-matched corr. on	perm. control on (naive)	perm. control on (f.-m.)
steels	5/5	5/5	5/5	5/5
expt_gap	5/5	5/5	5/5	5/5
expt_is_metal	1/5	5/5	5/5	5/5
glass	0/5	5/5	5/5	5/5

Table 5: Primary arm versus the Matbench v0.1 leaderboard (snapshot 18 October 2026). Classification: balanced accuracy at the threshold chosen inside each training fold, compared with the board's balanced-accuracy column. Rank counts our entry among the board entries.

Task	Metric	Ours: stack + naive corr.	Best board entry	Rank
steels	MAE	82.80 ± 11.65	TPOT-Mat: 79.95 ± 13.59	3 of 12
expt_gap	MAE	0.3240 ± 0.0140	Darwin: 0.2865 ± 0.0083	2 of 13
expt_is_metal	balanced acc.	0.9370 ± 0.0063	Darwin: 0.9598	2 of 8
glass	balanced acc.	0.8841 ± 0.0184	MODNet (v0.1.12): 0.9603	2 of 8

(5 parallel workers of 4 threads) one official fold took a median of about 1, 16, 8 and 44 minutes for steels, expt_gap, expt_is_metal and glass when every expert was computed from scratch (the glass figure includes a system-standby pause); the pair-excluded refits of the fold-matched arm account for 44%, 33% and 37% of the expert-fitting time on the first three tasks (84% on glass, inflated by the same pause).

Table 6: Calibration on the classification tasks (means over folds) and balanced accuracy at the inner-CV threshold.

Task	Arm	Brier	log loss	ECE	bal. acc.
expt_is_metal	Best single	0.0498	0.2007	0.0369	0.9378
expt_is_metal	Stack	0.0469	0.1662	0.0201	0.9374
expt_is_metal	Stack + naive corr.	0.0468	0.1662	0.0192	0.9370
expt_is_metal	Stack + fold-matched corr.	0.0469	0.1660	0.0211	0.9366
glass	Best single	0.0747	0.2451	0.0220	0.8827
glass	Stack	0.0735	0.2398	0.0211	0.8841
glass	Stack + naive corr.	0.0735	0.2398	0.0211	0.8841
glass	Stack + fold-matched corr.	0.0740	0.2411	0.0232	0.8835

Table 7: Mean simplex-stack weight per expert (over folds; "–" where the expert was not in the selected configuration of every fold).

Expert	steels	expt_gap	expt_is_metal	glass
et_all	0.94	0.86	0.41	0.57
krr_frac	0.00	0.00	–	–
krr_magpie	0.06	0.00	–	–
lgbm_all	0.00	0.14	0.48	0.27
lgbm_l1_all	–	–	–	–
mlp_all	0.00	0.00	0.05	0.00
rf_all	0.00	0.00	0.03	0.10
svc_frac	–	–	0.01	0.06
svc_magpie	–	–	0.02	0.00

S5. Run-2 arms, experts and stack weights

Table 8: Run-2 arms and experts on the official folds (mean ± s.d. over folds; probability ROC-AUC for classification).

Arm	steels MAE (MPa)	expt_gap MAE (eV)	expt_is_metal ROC-AUC	glass ROC-AUC
Best single (inner CV)	79.56 ± 9.90	0.2351 ± 0.0169	0.9900 ± 0.0010	0.9589 ± 0.0081
Equal-weight mean	95.18 ± 17.31	0.3172 ± 0.0112	0.9878 ± 0.0007	0.9524 ± 0.0095
Simplex stack	80.10 ± 10.19	0.2352 ± 0.0167	0.9895 ± 0.0008	0.9606 ± 0.0080
Final: stack + gated local corr.	80.10 ± 10.19	0.2352 ± 0.0167	0.9895 ± 0.0008	0.9606 ± 0.0080
Best single + gated corr.	79.56 ± 9.90	0.2351 ± 0.0169	0.9897 ± 0.0011	0.9589 ± 0.0081
Expert cat_F1	90.05	0.3075	0.9877	0.9475
Expert et_F1	84.47	0.2823	0.9856	0.9554
Expert et_F2	86.43	0.2819	0.9875	0.9600
Expert et_frac	79.56	–	0.9506	0.9476
Expert hurdle_et_F1_exp	–	0.2825	–	–
Expert hurdle_lgbm_F2_exp	–	0.2357	–	–
Expert hurdle_lgbm_F2_median	–	0.2360	–	–
Expert knn_frac	103.86	0.5019	0.9399	0.9078
Expert krr_frac	160.38	0.5613	–	–
Expert krr_magpie	109.37	0.4840	–	–
Expert lgbm_F2_tuned	88.38	0.2857	0.9900	0.9585
Expert mlp_F1	161.54	0.3950	0.9811	0.8937
Expert svc_frac	–	–	0.9472	0.8735
Expert svc_magpie	–	–	0.9566	0.9009
Expert xgb_F2	91.41	0.2902	0.9897	0.9571

S6. Ideation and prior-art process

Literature search, prior-art checking and idea generation for run 1 used the ResearchStudio Idea skills (paper-search, scoop-check, idea-spark), driven by an AI assistant. All records are in the repository (`research/`, `ideaspark_run/`).

- **Prior-art verdict: medium overlap.** Every component has ancestors (stacking, regression kriging, Δ-learning), but we found no work that combines them on these tasks.
- **Idea card.** idea-spark proposed the fold-matched correction (Section S2). Its coherence check verified the decomposition on synthetic instances.
- **Run outcome.** The run ended in a `needs_work` state because of an internal inconsistency in the tool's novelty gate. We therefore pre-registered the proposal only as a secondary arm.
- **Run-2 literature check.** A further search on DFT-informed and multi-fidelity prediction of experimental band gaps (`research/paper_search_dft/`) added the band-gap transfer and multi-fidelity references cited in the main text.

Table 9: Mean run-2 stack weight per expert over folds.

Expert	steels	expt_gap	expt_is_metal	glass
cat_F1	0.00	0.00	0.14	0.02
et_F1	0.00	0.00	0.00	0.03
et_F2	0.02	0.00	0.27	0.40
et_frac	0.85	–	0.01	0.24
hurdle_et_F1_exp	–	0.00	–	–
hurdle_lgbm_F2_exp	–	0.40	–	–
hurdle_lgbm_F2_median	–	0.60	–	–
knn_frac	0.03	0.00	0.01	0.00
krr_frac	0.00	0.00	–	–
krr_magpie	0.05	0.00	–	–
lgbm_F2_tuned	0.05	0.00	0.28	0.12
mlp_F1	0.00	0.00	0.07	0.00
svc_frac	–	–	0.00	0.00
svc_magpie	–	–	0.01	0.00
xgb_F2	0.00	0.00	0.21	0.19

S7. Leaderboard snapshot and label-based classification scores

The leaderboard values (`results/leaderboard_snapshot.json`) were taken from the per-task Matbench pages on 18 October 2026. For every listed classification entry the reported mean "ROC-AUC" equals the mean balanced accuracy to four decimals.

- Darwin's public results contain boolean labels.
- MODNet v0.1.12's contain probabilities, yet its board ROC-AUC equals its balanced accuracy. This is consistent with the older `matbench` scoring thresholding probabilities at 0.5.

We prepared two unsubmitted result files for each run: one with labels at the inner-CV threshold (comparable with the board) and one with probabilities.

S8. Post-hoc ablation: per-fold values and execution

The ablation (plan Section 12) was written after run 2 had been scored. It was not used for any selection, and it changes neither the reported pipeline nor the prepared leaderboard files.

- **Code.** Each variant re-runs the frozen run-2 code unchanged (`src/run_ablation.py`), with either the `dft_*` block or the three hurdle experts removed.
- **Cached experts.** Expert caches were reused only where the inputs were byte-identical:
  - in the no-DFT variant, the experts that read only Magpie or element-fraction columns;
  - in the no-hurdle variant, all remaining experts.
- **Interrupted run.** The run was interrupted once, when all workers were killed.
  - Before any scoring, the 16 completed folds were checked (loadable, finite, test identifiers equal to the official folds) and committed.

  - The 9 interrupted folds (glass 0–3, steels 0–4) were then rerun with the same launcher. Their partially written expert caches were checked to load and reused by the resumable pipeline. All experts are deterministic (deterministic LightGBM, fixed seeds).

- **Scoring.** All predictions were committed before the single scoring run (`src/evaluate_ablation.py`). It uses the same paired bootstrap as the run-2 evaluation, with no multiplicity adjustment because the ablation is descriptive.

Table 10: Per-fold board metric of the full run-2 pipeline and of each ablation variant (full / variant). For classification the mean probability ROC-AUC (full / variant) is also given. The last column gives the number of folds in which the variant's nested gate kept the local correction.

Task	Variant	fold 0	fold 1	fold 2	fold 3	fold 4	prob. ROC-AUC	corr. kept
expt_gap	no DFT block	0.2099 / 0.2582	0.2366 / 0.2765	0.2426 / 0.2744	0.2554 / 0.2713	0.2317 / 0.2816	–	0/5
expt_gap	no hurdle experts	0.2099 / 0.2624	0.2366 / 0.2789	0.2426 / 0.2853	0.2554 / 0.2908	0.2317 / 0.2770	–	0/5
expt_is_metal	no DFT block	0.9573 / 0.9441	0.9492 / 0.9400	0.9543 / 0.9461	0.9553 / 0.9471	0.9573 / 0.9351	0.9895 / 0.9849	0/5
steels	no DFT block	93.36 / 93.36	67.49 / 67.15	80.71 / 80.71	85.79 / 85.79	73.17 / 73.30	–	0/5
glass	no DFT block	0.9136 / 0.9110	0.8819 / 0.8845	0.8794 / 0.8766	0.9078 / 0.9024	0.8860 / 0.8863	0.9606 / 0.9608	0/5